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BUREAU OF AERONAUTICS
MILITARY SPECIFICATION

TRANSMITTING SET, RADIO AN/ART-28

This amendment shall form a part of and be attached to Specification MIL-T-6853 of 1 August 1950 and has been approved by the Bureau of Aeronautics.

Section 2

2.2 Amend this paragraph by making the following additions or changes:

Add "MIL-E-7894 - Electric Power, Aircraft, Characteristics of" to the list of specifications.

Change "AN-E-19" to "MIL-E-5400". (This change shall be made wherever it occurs in the specification. See paragraphs 3.1.1, 3.2.1, 3.2.6, 3.2.11, 3.2.13, 3.2.14, 4.1.2 and 4.2).

Delete "AN-C-176".

Change "RE-13A516" to "MIL-N-18307 - Nameplates and Nomenclature for Aircraft Electronic Equipment".

2.3 Amend by changing "NAVAER-2544" and "NAVAER-2586" to "MIL-D-18300 - Design Data Requirements for Contract Covering Aircraft Electronic Equipment".

2.4 Amend by deleting this paragraph.

Section 3

3.2.7 Amend by adding "except as modified below" to the end of this paragraph and adding the following new sub-paragraphs:

"3.2.7.1 Cooling - Although fans and/or blowers designed for a.c. operation are preferable, operation on d.c. is permissible in this equipment.

"3.2.7.2 Altitude - The equipment shall be required to operate only to an altitude of 30,000 feet.

"3.2.7.3 Sand and Dust - The equipment shall not be required to operate in a sand and dust atmosphere."

3.2.10 Amend by deleting this paragraph and substituting therefor the following:

"Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein.

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The power required shall not exceed the specified amounts.

(1) AC Power Three Phase

115/200 Volt Variable Frequency

Operating Voltage Limits	110 to 120 Volts
Operating Frequency Limits	320 to 1000 cps
Reduced Performance Voltage Limits	102-110 and 120-124 volts
Reduced Performance Frequency Limits	1 K V.A.
Power (3Ø)	

(2) DC Power 28 Volts System

Operating Voltage Limits	24 to 29 Volts
Power (Average)	140 watts
Maximum Surge Current	9 amps"

Amend by deleting the following paragraphs:

3.2.10.1, 3.2.10.2, 3.2.10.4 and 3.2.10.5. (These requirements are covered by Specification MIL-E-7894).

3.2.10.3 Amend by adding the following to this paragraph:

"Compliance with the requirements of this paragraph shall be determined by subjecting the equipment to operation with a power source within the voltage limits, at a frequency of 200 cycle per second at sea level, and at a temperature of 100°F for a period of 15 minutes".

3.2.10.6 Amend by changing "on Form NAVAER-2544" to "as required by MIL-E-7894". Also delete the last sentence.

3.3.1.1 Amend by deleting "and video-sync signal test generator" from the first sentence.

3.3.1.4 Amend by deleting this and the following paragraphs:

3.3.1.6 and 3.3.1.7.

3.3.1.8.1 Amend by changing from "200" to "100 watts."

3.3.1.8.2 Amend by changing from "800" to "400 watts."

3.3.1.9 Amend by changing from "ninety-five" to "eighty-five" and changing "four mc" to "two mc."

3.3.1.11.2 Amend by changing, in the last line, "55 db" to "50 db."

3.3.1.13 Amend by changing the second sentence to read as follows:

"The modulator shall have a video bandwidth such that the signal will not be attenuated by more than 3 db at 1.8 mc."

3.3.1.14 Amend by changing this paragraph to read as follows:

"Circuitry shall be provided to utilize a signal derived from the AC supply voltage to provide a video test signal in the absence of an external radar synchronizer."

3.3.1.16 Amend by deleting the "power ON-OFF switch, AN/ARR-27-AN/APS-20 input polarity switch", to "CW-MOD tuning switch".

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3.3.1.16.1 Amend by deleting this paragraph and substituting therefor the following:

"A meter shall be provided on the front panel to indicate power output to the antenna. This meter shall be so scaled to indicate subnormal performance, required CW power, and normal power output during modulation by incoming signals."

3.3.2.5 Amend by deleting "(3) negative 200 volts DC at 5 ma".

3.3.2.6 Amend by deleting this paragraph and substituting therefor the following:

"The operation of the power supply unit shall be automatically controlled by the operation of a relay in the transmitter unit. This relay shall operate a second relay in the power unit. The relay in the transmitter unit is operated by a remote ON-OFF switch. Both relays shall be actuated by 28 vdc from the transmitter unit."

3.3.2.7.2 Amend by changing "(2)" to read "Relay control voltage" instead of "negative 200 V DC".

Section 4

4.1.1 Amend by adding after this paragraph the following sub-paragraphs:

"4.1.1.1 Sand and dust requirements shall not apply to this equipment.

"4.1.1.2 The equipment shall be required to operate only to an altitude of 30,000 feet."

"4.1.1.3 Operation shall be checked after the equipment has been allowed a 15 minute warm-up time during which period no damage to the equipment has occurred."

Section 6

6.1 Amend by adding "400 cycle" after 115 volts.

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